NOLI · DATA ENGINEER BUSINESS CASE · SECTION 2

A self-service revenue platform finance can trust

Same architecture as Section 1. Different product: one definition of net revenue, fast dashboards, and Excel that cannot invent a second warehouse.

Audience: Senior Data Engineer · Principal Engineer 15 minutes
Built on: Fivetran · Databricks · Delta / Unity Catalog · Snowflake · dbt · Airflow · Power BI

Leadership wants to trust the numbers. The dashboard cannot explain itself.

Finance reports sales, refunds and discount impact from raw tables. Joins live in Power BI. 10–15s loads. Excel is the real BI tool.

Definition drift

Net revenue, GMV, discount rate and 'refunds' mean different things in three reports and a close spreadsheet.

Wrong grain in BI

Raw Fivetran orders ⋈ refunds ⋈ discounts in DAX. Every new analyst re-discovers fan-out bugs.

Latency as a trust issue

10–15s feels broken. People export and do not come back. Shadow numbers win.

No owner, little capacity

Pressure to 'just make it right' with a small data team. Self-serve must reduce interrupt load, not add a ticket queue.

Everyone can write a measure. That is the bug.

Metric	How it fragments today	Canonical rule (draft — Product + Finance sign-off)
Gross sales / GMV	With vs without tax; with vs without shipping; cancelled orders included.	Sum of fulfilled order lines, ex-VAT, before discounts, excluding cancelled / test / staff.
Discounts	Promo code only vs GWP vs employee vs shipping. Stacking unclear.	All commercial reductions at line grain, typed (code, automatic, GWP, staff). GWP is a discount type, not free revenue.
Refunds / returns	Partial refunds missed. Timing: order date vs refund date.	fct_refund at refund_date; net revenue restated for order_date and available as refund-period view.
Net revenue	GMV − discounts; or − refunds; or − both; or 'what Shopify said'.	GMV − discounts − refunds − chargebacks. Shipping collected is not net revenue unless Finance says so.
AOV / discount rate	Numerator and denominator from different filters.	Same order set. discount_rate = discounts / GMV. AOV = net / order_count.

We do not build a second platform. We add a finance product on Gold.

Keep	Add	Change	Defer
Fivetran RAW, hash vault, Airflow, Unity Catalog / Snowflake RBAC, dbt project, certified Power BI pattern.	Finance mart: fct_order_line, fct_refund, fct_discount, daily snapshot, metric layer. Semantic model. Analyze in Excel.	Ban BI on RAW. Split compute: dbt transform vs BI warehouse. Pre-aggregates for the exec pack. Metric tests.	Reverse ETL of 'VIP revenue' into CRM until identity + consent are clean. Real-time finance. A second BI tool.

The join moves left

Today: Power BI scans raw tables and joins them per visual.
Target: dbt incremental models produce a star schema + a daily finance agg. Power BI Import mode on that agg (and DirectQuery only if we must).
Attribution (Section 1) reads the same fct_order.net_revenue. One economic spine.

NOLI · Self-service finance platform on the same architecture

One net_revenue · line grain for truth · day grain for speed · Excel cannot invent a second warehouse

Same ingest
Fivetran RAW already landing

Same hash vault
Hashed customer_sk only

Same Airflow
Last-good if tests fail

New finance mart
dbt Gold product

Certified dataset
Power BI + live Excel

Metric catalogue
Owner + definition + as-of

TODAY · THE FAILURE MODE

Raw tables in Power BI
Fivetran orders ▯ refunds ▯ discounts in DAX
Fan-out, 10–15s loads, three definitions of net revenue

Excel as a warehouse
Exports drift from the dashboard
Close pack cannot be explained

No owner of the formula
Leadership pressure, small data team
Every analyst "fixes" it locally

What we stop
New datasets on RAW
DAX that rewrites net revenue
Silent VAT / GWP / staff orders
Genie or AI on uncertified tables

TARGET · FINANCE PRODUCT ON GOLD

SAME SPINE AS SECTION 1

fct_order_line
Atomic commercial fact · SKU, brand, GMV, discount, net, tax · header = sum(lines) test

fct_discount
Type: code / auto / GWP / staff
funded_by: Noli vs brand

fct_refund
Partial/full · refund_date
Restates net on order_date too

agg_finance_day
date × brand × channel × discount_type · pre-summed for p95 < 3s exec view

net_revenue = gmv − discounts − refunds − chargebacks
ex-VAT, GBP · exclude test / staff / cancelled · Finance + DPM sign the words
Implemented once in dbt · Power BI is SUM(net_revenue) · Excel live on the same dataset

HOW ANALYSTS WORK

Find
Endorsed "Certified — Finance" dataset
One-page metric catalogue

Explore
Date, brand, channel, discount type
No warehouse keys on day one

Ask for more
New metric = dbt PR, not a CSV column
Trivial metric SLA: 2–3 days

Understand
Hover: definition + as-of time
Lineage to the dbt model
Last-good if conservation fails

SUCCESS

- p95 < 3s exec view
- Tickets −70%
- 100% close packs certified
- 0 RAW-bound finance models
- 14d tests green
- ≤ 3d new metric
- Attribution $ = finance $

Keep from S1
Fivetran, hash vault, Airflow,
Unity Catalog, dbt project

Add
Finance mart, day agg,
semantic model, Excel live

Change
Ban BI on RAW · split compute
Pre-aggregates for the pack

Defer
Reverse ETL · real-time finance
Second BI tool

Privacy
No email in the dataset
Same hashed customer_sk

AI / JD projects
Features from fct_order_line
Clean room = brand sell-out

Section 2 does not build a second platform. The join moves left. Meaning stays in dbt. Power BI and Excel only consume.

Line grain for truth. Day grain for speed. Metrics for language.

fct_order_line

SKU, brand, category, qty, gmv, discount, net, tax, shipping share. The atomic commercial fact.

fct_order

Header rollup of the same rules. Status, customer_sk (hashed), channel of order — not attribution credit.

fct_discount

One row per discount application. Type, code, funded_by (Noli vs brand).

fct_refund

Partial/full, reason, refund_ts, original_order_id, amount. Restates net.

agg_finance_day

date × brand × channel × discount_type. Pre-summed for the 3-second dashboard.

metric_* / semantic

dbt metrics or a thin semantic layer: net_revenue, gmv, discount_rate, refund_rate, aov.

dim_product

Brand, hero category, margin class if we have it. SCD2 for price changes if needed.

dim_customer

Same hashed key as attribution. No PII. Cohort month. Consent flags.

One formula, three interfaces — warehouse, BI, Excel

Draft metric · owned by Finance + Data Product · implemented in dbt

net_revenue = gmv − discounts − refunds − chargebacks (ex-VAT, GBP, exclude test/staff/cancelled)

dbt model + test

Column net_revenue on fct_order_line and fct_order. Test: header = sum(lines). Test: refunds cannot exceed original net.

Metric layer

Named metric with dimensions (date, brand, channel, discount_type). Versioned in git. Shown in dbt docs.

Power BI dataset

Thin measures: SUM(net_revenue). No rewrite of the formula. Display folders and descriptions from the catalogue.

Excel

Analyze in Excel / live connection to the certified dataset. Pivot only. If they need a new metric, it is a dbt PR — not a new column in a CSV.

10–15 seconds is a modelling problem wearing a BI costume

Today	Why it hurts	Change
DirectQuery / live on raw Fivetran tables	Wide JSON-ish tables, no cluster key the BI query hits, joins per visual.	Import (or composite) on agg_finance_day + a few dims. RAW hidden.
Joins in the model for every refund	Fan-out, duplicate GMV, unpredictable totals.	Pre-joined facts in dbt. BI relates on surrogate keys only.
No incremental mart	Every refresh rebuilds history. Warehouse and dataset fight.	Incremental dbt + incremental Power BI refresh on date.
One warehouse for everything	A backfill evicts the cache; finance waits.	Dedicated BI warehouse/SQL endpoint, auto-resume, small.
Too many visuals, too many slicers	Each slicer = another scan.	Exec pack: 6 visuals, date + brand + channel. Detail page optional.
Wide text columns in memory	Import mode bloated; refresh fails.	Mart is typed, hashed IDs, no PII, no raw payloads.

Self-serve is a product: certified data, a catalogue, and a paved road

Find

dbt docs + exposures. A one-page metric catalogue (name, grain, owner, last certified). Power BI endorsed dataset only in the finance workspace.

Explore

Shared semantic model: date, brand, channel, discount type. Analysts build reports from the dataset — they do not get warehouse keys on day one.

Excel, safely

Analyze in Excel / live pivot on the certified model. Monthly close can still be a workbook — it cannot be a private extract of RAW.

Ask for more

New metric = issue → dbt PR → test → release notes. SLA: trivial metric 2–3 days, not a 6-week project. Data Product Manager prioritises.

Understand

Every official visual has a hover: definition + as-of timestamp + default filters. Lineage link to the dbt model. No mystery tiles.

Don't

A second unofficial dataset 'just for this deck'. Personal gateways. Copy-paste of net revenue DAX. Sharing RAW via Teams.

Guardrails, not gatekeeping

Access & privacy

- Entra groups: finance_read, marketing_read, de_admin.
- Row-level: not required for UK-only start; ready for region later.
- Column-level: no email, phone, scan IDs in the finance dataset.
- Same hash keys as Section 1 — finance can cohort, not identify.
- Unity Catalog + Snowflake grants versioned (Terraform or GRANT models).
- Workspace endorsement: only 'Certified — Finance' is used in exec packs.
- Audit: who refreshed, who exported. Exports of line-level get a warning.

Change control

- Metric changes are PRs with Finance as reviewer.
- Breaking change (net revenue formula) = versioned metric + dual-run.
- dbt CI on PRs: tests must pass before merge.
- Airflow blocks Power BI refresh if conservation tests fail — we serve last-good.
- Release note in Teams/Slack: what changed, who signed it.
- Quarterly: kill unused datasets (Power BI lineage + dbt exposures).
- Cost: BI warehouse budget; no 24/7 XL cluster for a daily pack.

Proactive reliability — the JD — without a 20-person platform group

Data Product Manager	Priority, definition workshops, 'this is the number we will defend'.
Data Tech Lead	Standards, stack boundaries, review of dbt contracts and Airflow SLAs.
Data Engineer (this role)	Pipelines, tests, marts, performance, hash/RTBF, on-call for freshness.
Finance / Marketing	Sign definitions. Stop screenshots of Ads Manager as source of truth.
AI / ML stakeholders	Read feature tables from Gold, not from finance Excel. Same spine.

Reverse ETL is in the JD. I would not start there.

Later

Push LTV, predicted churn, or 'high discount reliance' to CRM / email — from certified Gold, hashed identity, consent-checked. Tool: existing reverse ETL or a thin job. Same Airflow DAG family.

Not now

Syncing raw emails or half-correct revenue into Klaviyo will industrialise the inconsistency we are here to kill. Activation without a metric layer is how ops and finance diverge forever.

Fit to Noli example projects

Traffic analysis — onsite tracking quality feeds the same UTM + identity spine; finance does not wait on it.
Business data availability — this mart is the deliverable.
Clean room — share brand-level sell-out / campaign performance, not customer lists.
AI enablement — feature tables (discount sensitivity, repeat purchase) built on fct_order_line, not on a dashboard extract.

If we cannot measure it, it is a rebrand of the same mess

< 3s

p95 load, exec revenue view

≥ 70%

drop in 'numbers don't match' tickets

100%

board / close packs on certified dataset

0

Power BI models still bound to RAW

14d

conservation + freshness tests green

≤ 3d

time-to-ship a trivial new metric

Sequence: stop the bleeding, certify one number, then open self-serve

Days 1–30

Define & protect

Workshop net revenue with Finance + DPM.
Hide RAW from the finance workspace.
Ship fct_order_line + header conservation test.
Performance baseline (analyzer + query history).
Last-good refresh policy in Airflow.

Days 31–60

Make it fast and official

agg_finance_day + Import-mode certified dataset.
Exec dashboard v1: sales, refunds, discount impact.
Analyze in Excel on that dataset only.
Metric catalogue (one page is enough).
p95 under 3s or we keep cutting the model.

Days 61–90

Self-serve without chaos

Analyst workspace on the certified model.
Request SLA for new metrics.
Kill or quarantine unofficial reports.
Hook attribution net to finance net (recon).
Write the runbook; measure ticket drop.

Downsides, said out loud

dbt + Power BI overlap

If someone 'quickly' rebuilds net revenue in DAX, we are back to day zero. Mitigation: dataset permissions + review of published measures.

Two warehouses

Snowflake + Databricks without a Gold SoR doubles cost and definitions. Mitigation: one serving layer for finance.

Fivetran bill

Finance wants every connector 'just in case'. Mitigation: model-driven connectors; pause unused MAR.

Over-modelling

A 40-dim Kimball cathedral delays the close. Mitigation: line fact + 5 dims, then grow.

AI on junk

Genie/Chat on RAW will sound confident and be wrong. Mitigation: AI only on certified Gold.

People

Self-serve fails if Finance never attends the definition workshop. Mitigation: DPM-owned decision log.

CLOSE

The platform in one sentence

Ingest with Fivetran, process the hard things in Databricks, govern with Delta / Unity Catalog, serve from Snowflake, define meaning in dbt, run the clock with Airflow, and let Power BI and Excel consume — never redefine — a hashed, tested Gold model.

Trust

One net revenue. Conservation tests. Last-good publish.

Speed

Day-level mart. Import mode. Separate BI compute.

Ease

Certified dataset, catalogue, Excel live, metric SLA.

I would start Monday with the definition workshop and turning off RAW access in the finance workspace. Everything else is implementation.